Lab – File System Commands

Introduction

In this lab, you will use CLI commands to manage files and folders in Windows.

Recommended Equipment

- A computer running Windows

Instructions

Step 1: Access the Windows command prompt.

a. Log on to a Windows computer. The account **ITEUser** is used as the example user account throughout this lab.

b. To access the Windows command prompt, click **Start** and enter **cmd**. Select **Command Prompt** or **cmd.exe** to continue.

Step 2: Create and change directories.

In this step, you will use the change directory (**cd**), make directory (**md**), and directory (**dir**) commands.

Note: A directory is another word for folder. Directory and folder are used interchangeably throughout this lab.

a. Type **cd** at the command prompt.

Question:

What is the current directory?

```
C:\Users\TIPQC>MD.CPE11S4CAMBEL

C:\Users\TIPQC>cd
C:\Users\TIPQC
```

b. Type **dir** at the command prompt to list the files and folders that are in the current folder.

```
C:\Users\ITEUser> dir
 Volume in drive C has no label.
 Volume Serial Number is 9055-35E9

 Directory of C:\Users\ITEUser

04/27/2019  09:21 AM    <DIR>          .
04/27/2019  09:21 AM    <DIR>          ..
04/25/2019  11:39 AM    <DIR>          3D Objects
04/25/2019  11:39 AM    <DIR>          Contacts
04/26/2019  10:29 AM    <DIR>          Desktop
04/25/2019  11:39 AM    <DIR>          Documents
```

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```
04/25/2019  11:39 AM    <DIR>          Downloads
<some output omitted>
               0 File(s)              0 bytes
              15 Dir(s)  32,671,969,280 bytes free
```

c. In the current directory, use the **md** command to create three new folders: **ITEfolder1**, **ITEfolder2**, and **ITEfolder3**. Type **md ITEfolder1** and press **Enter**. Create **ITEfolder2** and **ITEfolder3**. (**Note**: The command **mkdir** performs the same function as the command **md**.)

```
C:\Users\ITEUser> md ITEfolder1
C:\Users\ITEUser> md ITEfolder2 ITEfolder3
```

```
10/08/2026  11:24 AM    <DIR>          .CPE11S4CAMBEL
10/08/2026  11:41 AM    <DIR>          .CPE11S4CAMBEL1
10/08/2026  11:41 AM    <DIR>          .CPE11S4CAMBEL2
10/08/2026  11:41 AM    <DIR>          .CPE11S4CAMBEL3
```

d. Type **dir** to verify that the folders have been created.

e. Type **cd ITEfolder3** at the command prompt and press **Enter**.

Question:

Which folder are you in now?

```
C:\Users\TIPQC>cd.CPE11S4CAMBEL

C:\Users\TIPQC\.CPE11S4CAMBEL>CD.CPE11S4CAMBEL3
The system cannot find the path specified.

C:\Users\TIPQC\.CPE11S4CAMBEL>CD
C:\Users\TIPQC\.CPE11S4CAMBEL

C:\Users\TIPQC\.CPE11S4CAMBEL>cd
C:\Users\TIPQC\.CPE11S4CAMBEL
```

f. Create **ITEfolder4** in the current directory. Within the **ITEfolder4** folder, create a folder named **ITEfolder5**. Use the **dir** command to verify the folder creation.

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Question:

What command or commands did you use to create the nested folders?

```
C:\Users\TIPQC\.CPE11S4CAMBEL\.CPE11S4CAMBEL4>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL\.CPE11S4CAMBEL4

10/08/2026  11:52 AM    <DIR>          .
10/08/2026  11:52 AM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL5
               0 File(s)              0 bytes
               3 Dir(s)  505,705,308,160 bytes free
```

g. Change directory as necessary until you are in ITEfolder5.

h. Type **cd ..** to change the current directory. Each **..** is a shortcut to move up one level in the directory tree.

Question:

After issuing the **cd ..** command, what is your directory now?

```
C:\Users\TIPQC\.CPE11S4CAMBEL3\.CPE11S4CAMBEL4\.CPE11S4CAMBEL5>cd..
```

What would be the current directory if you issue this command at C:\Users\ITEUser\ITEfolder3\ITEFolder4?

```
C:\Users\TIPQC\.CPE11S4CAMBEL3\.CPE11S4CAMBEL4>
```

Step 3: Create text files.

i. Navigate to the **C:\Users\ITEUser\ITEfolder1** directory. Type **cd ..\ITEfolder1** at the prompt.

```
C:\Users\TIPQC\.CPE11S4CAMBEL>
```

j. Type **echo This is doc1.txt > doc1.txt** at the command prompt. The **echo** command is used to display a message at the command prompt. The **>** is used to redirect the message from the screen to a file. For example, in the first line, the message **This is doc1.txt** is redirected into a new file named **doc1.txt**. Use the **echo** command and **>** redirect to create these files: **doc2.txt**, **file1.txt**, and **file2.txt**.

```
C:\Users\ITEUser\ITEfolder1> echo This is doc1.txt > doc1.txt
C:\Users\ITEUser\ITEfolder1> echo This is doc2.txt > doc2.txt
C:\Users\ITEUser\ITEfolder1> echo This is file1.txt > file1.txt
C:\Users\ITEUser\ITEfolder1> echo This is file2.txt > file2.txt
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>echo
ECHO is on.
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL>echo This is doc.1 txt > doc.txt

C:\Users\TIPQC\.CPE11S4CAMBEL>echo This is doc.2 txt > doc2.txt

C:\Users\TIPQC\.CPE11S4CAMBEL>echo This is file1.txt > file1.txt

C:\Users\TIPQC\.CPE11S4CAMBEL>echo This is file2.txt > file2.txt
```

k. Use the **dir** command to verify the files are in the **ITEfolder1** folder.

```
C:\Users\ITEUser\ITEFolder1> dir
<some output omitted>
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               4 File(s)             78 bytes
               2 Dir(s)  32,625,397,760 bytes free
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>dir
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:11 PM    <DIR>          .
10/08/2026  11:41 AM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:11 PM                20 doc2.txt
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               4 File(s)             80 bytes
               3 Dir(s)  505,069,174,784 bytes free
```

l. Use the **more** or **type** command to view the content of the newly created text files.

```
C:\Users\ITEUser\ITEfolder1> more doc1.txt
This is doc1.txt

C:\Users\ITEUser\ITEfolder1> type doc2.txt
This is doc2.txt
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL>type doc2.txt
This is doc.2 txt

C:\Users\TIPQC\.CPE11S4CAMBEL>more doc.txt
This is doc.1 txt
```

Step 3: Copy, delete, and move files.

a. At the command prompt, type **move doc2.txt C:\Users\ITEUser\ITEfolder2** to move the file **doc2.txt** to the **C:\Users\ITEUser\ITEfolder2** directory.

```
C:\Users\ITEUser\ITEfolder1> move doc2.txt C:\Users\ITEUser\ITEfolder2
        1 file(s) moved.
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>move doc2.txt C:\Users\TIPQC\CPE11S4CAMBEL2
        1 file(s) moved.
```

b. Type **dir** at the prompt to verify that **doc2.txt** is no longer in the current directory.

```
C:\Users\TIPQC\.CPE11S4CAMBEL>dir
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:18 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               3 File(s)             60 bytes
               3 Dir(s)  505,051,099,136 bytes free
```

c. Navigate to **C:\Users\ITEUser\ITEfolder2** to change the directory to **ITEfolder2**. Type **dir** at the prompt to verify **doc2.txt** has been moved.

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```
C:\Users\TIPQC\.CPE11S4CAMBEL2>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL2

10/08/2026  11:41 AM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  505,082,314,752 bytes free
```

d. Type **copy doc2.txt doc2_copy.txt** to create a copy of **doc2.txt**. Type **dir** at the prompt to verify a copy of the file has been created.

```
C:\Users\ITEUser\ITEfolder2> dir
<some output omitted>
Directory of C:\Users\ITEUser\ITEfolder2

04/30/2019  09:07 AM                19 doc2.txt
04/30/2019  09:07 AM                19 doc2_copy.txt
               2 File(s)             38 bytes
               2 Dir(s)  31,753,068,544 bytes free
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>dir
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:30 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               3 File(s)             60 bytes
               3 Dir(s)  501,289,455,616 bytes free
```

e. Use the **move** command to move **doc2_copy.txt** to **ITEfolder1**. Type **move doc2_copy.txt ..\ITEfolder1**.

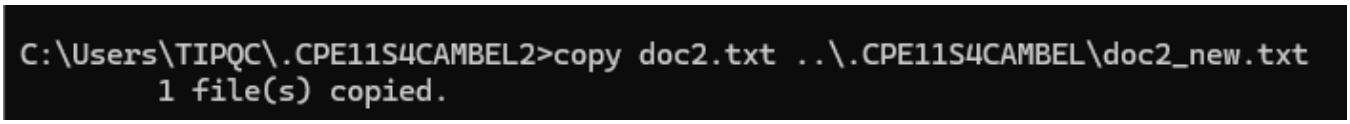
```
C:\Users\ITEUser\ITEfolder2> move doc2_copy.txt ..\ITEfolder1
        1 file(s) moved.
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL2>move doc2_copy.txt ..\.CPE11S4CAMBEL
        1 file(s) moved.
```

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f. A copy of the file **doc2.txt** can be created in a new location using the **copy** command. At the prompt, enter the **copy doc2.txt ..\ITEfolder1\doc2_new.txt**.

```
C:\Users\ITEUser\ITEfolder2> copy doc2.txt ..\ITEfolder1\doc2_new.txt
        1 file(s) copied.
```



g. The file **doc2.txt** can also be moved to a new location with a new filename using the **move** command. Type **move doc2.txt ..\ITEfolder1\doc2_new.txt** at the prompt.

```
C:\Users\ITEUser\ITEfolder2> move doc2.txt ..\ITEfolder1\doc2_move.txt
        1 file(s) moved.
```



h. Type **dir ..\ITEfolder1** to view the content in **ITEfolder1** without leaving the current directory.

```
C:\Users\ITEUser\ITEfolder2> dir ..\ITEfolder1
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1

04/29/2019  12:08 PM    <DIR>          .
04/29/2019  12:08 PM    <DIR>          ..
04/29/2019  08:05 AM                19 doc1.txt
04/29/2019  08:06 AM                19 doc2_copy.txt
04/29/2019  08:06 AM                19 doc2_move.txt
04/29/2019  08:06 AM                19 doc2_new.txt
04/29/2019  08:08 AM                20 file1.txt
04/29/2019  08:08 AM                20 file2.txt
               6 File(s)            116 bytes
               2 Dir(s)  31,467,700,224 bytes free
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL2>DIR..\.CPE11S4CAMBEL
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:42 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:28 PM                19 doc2_copy.txt
10/08/2026  12:28 PM                19 doc2_move.txt
10/08/2026  12:28 PM                19 doc2_new.txt
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               6 File(s)            117 bytes
               3 Dir(s)  504,845,434,880 bytes free
```

i. Change the current directory to **ITEfolder1**. Type **cd ..\ITEfolder1** at the prompt.

```
C:\Users\TIPQC\.CPE11S4CAMBEL2>CD..

C:\Users\TIPQC>CD.CPE11S4CAMBEL
```

j. Move **file1.txt** and **file2.txt** into **ITEfolder3**. To move all the files that contain the word **file** into **ITEfolder3** with one command, use a **wildcard** (*) character to represent one or more characters. Type **move file* ..\ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> move file* ..\ITEfolder3
C:\Users\ITEUser\ITEfolder1\file1.txt
C:\Users\ITEUser\ITEfolder1\file2.txt
        2 file(s) moved.
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>move file* ..\.CPE11S4CAMBEL3
C:\Users\TIPQC\.CPE11S4CAMBEL\file1.txt
C:\Users\TIPQC\.CPE11S4CAMBEL\file2.txt
        2 file(s) moved.
```

k. To view the content of the file doc2_new.txt, use the **type** or **more** command at the prompt.

```
C:\Users\ITEUser\ITEfolder1> type doc2_new.txt
This is doc2.txt
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>type doc2_new.txt
This is doc2.txt
```

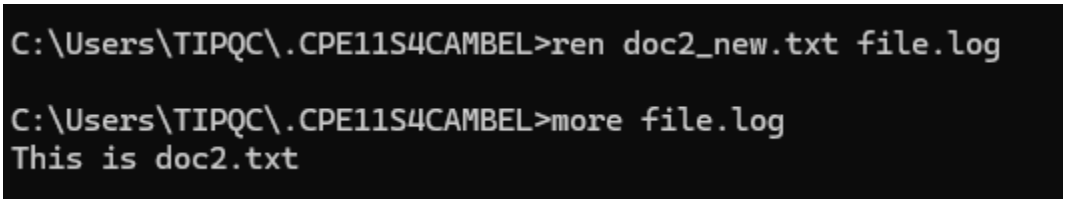
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l. Use the **ren** command to rename doc2_new.txt to doc3.txt. Use the **dir** command to display the content in the directory.

```
c:\Users\ITEUser\ITEfolder1> ren doc2_new.txt file.log
```

m. Use the **type** or **more** command to view the content of the newly rename file **file.log.**

```
C:\Users\ITEUser\ITEfolder1> more file.log
This is doc2.txt
```



n. To delete a file or multiple files, use the **del** command with the file names at the prompt.

Question:

What single command would you use to delete all the files with doc2 in the filename? Use the command to remove the files with doc2 in the filename.

Type your answers here.

What command would you use to delete all the files in the directory? Use the command to remove the files.

Type your answers here.

Step 4: Use the xcopy and robocopy commands.

In this step, the **xcopy** and **robocopy** commands are used to copy the content in a directory.

a. View the content of ITEfolder3.

```
C:\Users\ITEUser\ITEfolder1> dir ..\ITEfolder3
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder3

04/29/2019  03:11 PM    <DIR>          .
04/29/2019  03:11 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  03:01 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,492,157,440 bytes free
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL>dir ..\.CPE11S4CAMBEL3
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL3

10/08/2026  12:46 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  12:02 PM    <DIR>          .CPE11S44
10/08/2026  12:04 PM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               2 File(s)             40 bytes
               4 Dir(s)  504,914,259,968 bytes free
```

b. Type **xcopy ..\ITEfolder3 .** at the prompt to copy the content of **ITEfolder3** to **ITEfolder1**. Note the **.** at the end of the command. It is a shortcut for the current directory.

```
C:\Users\ITEUser\ITEfolder1> xcopy ..\ITEfolder3 .
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>xcopy ..\.CPE11S4CAMBEL3 .
..\.CPE11S4CAMBEL3\file1.txt
..\.CPE11S4CAMBEL3\file2.txt
2 File(s) copied
```

c. At the prompt, type **dir** to display the content of **ITEfolder1**. Only the files in the **ITEfolder3** were copied into **ITEfolder1**. The directory **ITEfolder4** was not copied into **ITEfolder3**.

```
C:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
 Directory of C:\Users\ITEUser\ITEfolder1
04/29/2019  03:16 PM    <DIR>          .
04/29/2019  03:16 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
               2 File(s)             40 bytes
               2 Dir(s)  31,491,321,856 bytes free
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:56 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  11:52 AM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:28 PM                19 doc2_copy.txt
10/08/2026  12:28 PM                19 doc2_move.txt
10/08/2026  12:28 PM                19 file.log
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               6 File(s)            117 bytes
               3 Dir(s)  504,913,858,560 bytes free
```

d. Use **help xcopy** or **xcopy /?** to determine which switch would allow the **xcopy** command to copy **all** the files and directories.

Question:

What option allows you to copy all the files and directories, including the empty directories?

Type your answers here.

e. Because **ITEfolder4** is a subfolder and **ITEfolder5** is both a subfolder and an empty folder, **/E** is needed to copy all the contents of **ITEfolder3** and the empty subfolder.

Type **xcopy /E ..\ITEfolder3 .** at the prompt to copy the files. When prompted, type **a** to allow overwriting the existing files.

```
c:\Users\ITEUser\ITEfolder1> xcopy /E ..\ITEfolder3
Overwrite C:\Users\ITEUser\ITEfolder1\file1.txt (Yes/No/All)? a
..\ITEfolder3\file1.txt
..\ITEfolder3\file2.txt
2 File(s) copied
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>xcopy /E ..\.CPE11S4CAMBEL3
Overwrite C:\Users\TIPQC\.CPE11S4CAMBEL\file1.txt (Yes/No/All)? a
..\.CPE11S4CAMBEL3\file1.txt
..\.CPE11S4CAMBEL3\file2.txt
2 File(s) copied
```

f. Verify that the **ITEfolder4** and **ITEfolder5** were also copied in **ITEfolder1**.

```
c:\Users\ITEUser\ITEfolder1> dir
<some output omitted>
```

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```
 Directory of c:\Users\ITEUser\ITEfolder1

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:05 PM                20 file1.txt
04/29/2019  03:05 PM                20 file2.txt
04/29/2019  04:41 PM    <DIR>          ITEfolder4
               2 File(s)             40 bytes
               3 Dir(s)  31,493,193,728 bytes free
```

```
C:\Users\TIPQC\.CPE11S4CAMBEL>
C:\Users\TIPQC\.CPE11S4CAMBEL>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL

10/08/2026  12:57 PM    <DIR>          .
10/08/2026  12:18 PM    <DIR>          ..
10/08/2026  12:02 PM    <DIR>          .CPE11S44
10/08/2026  12:04 PM    <DIR>          .CPE11S4CAMBEL4
10/08/2026  12:10 PM                20 doc.txt
10/08/2026  12:28 PM                19 doc2_copy.txt
10/08/2026  12:28 PM                19 doc2_move.txt
10/08/2026  12:28 PM                19 file.log
10/08/2026  12:11 PM                20 file1.txt
10/08/2026  12:11 PM                20 file2.txt
               6 File(s)            117 bytes
               4 Dir(s)  504,902,692,864 bytes free
```

```
c:\Users\ITEUser\ITEfolder1> dir ITEfolder4
<some output omitted>
 Directory of c:\Users\ITEUser\ITEfolder1\ITEfolder4

04/29/2019  04:41 PM    <DIR>          .
04/29/2019  04:41 PM    <DIR>          ..
04/29/2019  03:00 PM    <DIR>          ITEfolder5
               0 File(s)              0 bytes
               3 Dir(s)  31,493,193,728 bytes free
```

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```
C:\Users\TIPQC\.CPE11S4CAMBEL>dir .CPE11S4CAMBEL4
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL\.CPE11S4CAMBEL4

10/08/2026  12:04 PM    <DIR>          .
10/08/2026  12:57 PM    <DIR>          ..
10/08/2026  12:04 PM    <DIR>          .CPE11S4CAMBEL5
               0 File(s)              0 bytes
               3 Dir(s)  504,901,918,720 bytes free
```

g. The robocopy command can also be used to copy the content of a directory to a new destination. The robocopy command has more capabilities than the built-in Windows copy and xcopy command, such as resume copying after a network interruption, skip files that appear to be identical to the files in the destination folders, and mirror a directory by keeping the destination directory in sync with the source directory.

Copy the ITEfolder4 content to ITEUser folder using the robocopy command.

```
C:\Users\ITEUser\ITEfolder1> robocopy /E
C:\Users\ITEUser\ITEfolder3\ITEfolder4\ C:\Users\ITEUser
```

Note the information provided by the command during the copying process.

h. Navigate to **C:\Users\ITEUser** to verify that the folder ITEfolder5 is copied.

```
C:\Users\TIPQC\.CPE11S4CAMBEL3\.CPE11S4CAMBEL4>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC\.CPE11S4CAMBEL3\.CPE11S4CAMBEL4

10/08/2026  12:04 PM    <DIR>          .
10/08/2026  12:46 PM    <DIR>          ..
10/08/2026  12:04 PM    <DIR>          .CPE11S4CAMBEL5
               0 File(s)              0 bytes
               3 Dir(s)  504,934,305,792 bytes free
```

Step 5: Delete directories.

In this step, you will delete an empty and a non-empty directory using the **rd** command.

a. Use the **rd ITEfolder2** to delete the empty directory and verify that the directory was deleted.

```
C:\Users\ITEUser> rd ITEfolder2
```

b. Navigate to the **C:\Users\ITEUser\ITEfolder3** directory.

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```
 Directory of C:\Users\TIPQC

10/08/2026  01:06 PM    <DIR>          .
03/12/2026  01:42 PM    <DIR>          ..
02/14/2025  02:28 PM    <DIR>          .anaconda
03/19/2025  06:09 PM    <DIR>          .azuredatastudio
11/12/2025  10:57 AM    <DIR>          .cache
04/02/2025  02:04 PM    <DIR>          .conda
02/14/2025  02:24 PM               146 .condarc
02/14/2025  02:23 PM    <DIR>          .continuum
08/15/2026  01:44 PM    <DIR>          .copilot
10/08/2026  12:57 PM    <DIR>          .CPE11S4CAMBEL
10/08/2026  11:41 AM    <DIR>          .CPE11S4CAMBEL1
10/08/2026  12:46 PM    <DIR>          .CPE11S4CAMBEL3
07/18/2026  02:55 PM    <DIR>          .ipynb_checkpoints
02/05/2025  03:24 PM    <DIR>          .ipython
02/14/2025  02:28 PM    <DIR>          .jupyter
02/05/2025  03:24 PM    <DIR>          .matplotlib
10/21/2024  01:23 PM               176 .packettracer
07/18/2026  04:31 PM    <DIR>          .spyder-py3
10/08/2026  12:37 PM    <DIR>          .VirtualBox
02/04/2025  10:51 AM    <DIR>          .vscode
07/29/2026  05:48 PM    <DIR>          .vscode-shared
01/29/2026  02:34 PM                 0 30
02/14/2025  02:27 PM    <DIR>          anaconda_projects
10/21/2024  01:24 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
10/08/2026  12:11 PM                20 CPE11S4CAMBEL2
07/30/2026  10:47 AM    <DIR>          Desktop
10/07/2026  05:48 PM    <DIR>          Documents
10/08/2026  11:34 AM    <DIR>          Downloads
04/07/2025  02:21 PM    <DIR>          Favorites
10/22/2025  05:58 PM                 6 jcbd.txt
10/22/2025  05:56 PM                 2 jcd.txt
02/05/2025  03:29 PM    <DIR>          Jedi
10/22/2025  06:13 PM    <DIR>          journals
10/18/2024  05:21 AM    <DIR>          Links
10/22/2025  05:53 PM                 2 m.txt
10/07/2026  05:04 PM    <DIR>          Microsoft
10/18/2024  05:21 AM    <DIR>          Music
10/08/2026  10:30 AM    <DIR>          OneDrive
01/20/2025  03:16 PM    <DIR>          Pictures
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
02/14/2025  02:28 PM                72 Untitled.ipynb
02/14/2025  02:29 PM                72 Untitled1.ipynb
03/21/2025  02:42 PM             2,297 Untitled10.ipynb
04/04/2025  02:28 PM             1,913 Untitled11.ipynb
04/25/2025  03:03 PM               628 Untitled12.ipynb
07/18/2026  02:55 PM               337 Untitled13.ipynb
02/14/2025  02:53 PM             2,711 Untitled2.ipynb
02/14/2025  03:11 PM                72 Untitled3.ipynb
02/14/2025  03:16 PM             3,904 Untitled4.ipynb
02/28/2025  02:32 PM               628 Untitled5.ipynb
02/28/2025  03:28 PM                72 Untitled6.ipynb
03/07/2025  03:33 PM               804 Untitled7.ipynb
03/07/2025  03:58 PM             2,334 Untitled8.ipynb
03/07/2025  04:00 PM             2,989 Untitled9.ipynb
08/06/2025  05:04 PM    <DIR>          venv
10/21/2024  01:20 PM    <DIR>          Videos
10/08/2026  11:23 AM    <DIR>          VirtualBox VMs
              21 File(s)         19,185 bytes
              38 Dir(s)  504,927,379,456 bytes free
```

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c. Use the **rd ITEfolder1** to delete the directory. Verify the directory removal using the **dir** command.

Were you able to delete the directory? Explain.

Type your answers here.

d. Use **rd /?** command to determine the switch that allows the deletion of a non-empty directory.

```
c:\Users\ITEUser\ITEfolder3> rd /S ITEfolder1
ITEfolder4, Are you sure (Y/N)? y
```

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```
C:\Users\TIPQC>RD /S .CPE11S4CAMBEL
.CPE11S4CAMBEL, Are you sure (Y/N)? Y

C:\Users\TIPQC>DIR
 Volume in drive C is Q3203-20
 Volume Serial Number is 916B-E0A9

 Directory of C:\Users\TIPQC

10/08/2026  01:12 PM    <DIR>          .
03/12/2026  01:42 PM    <DIR>          ..
02/14/2025  02:28 PM    <DIR>          .anaconda
03/19/2025  06:09 PM    <DIR>          .azuredatastudio
11/12/2025  10:57 AM    <DIR>          .cache
04/02/2025  02:04 PM    <DIR>          .conda
02/14/2025  02:24 PM               146 .condarc
02/14/2025  02:23 PM    <DIR>          .continuum
08/15/2026  01:44 PM    <DIR>          .copilot
10/08/2026  12:46 PM    <DIR>          .CPE11S4CAMBEL3
07/18/2026  02:55 PM    <DIR>          .ipynb_checkpoints
02/05/2025  03:24 PM    <DIR>          .ipython
02/14/2025  02:28 PM    <DIR>          .jupyter
02/05/2025  03:24 PM    <DIR>          .matplotlib
10/21/2024  01:23 PM               176 .packettracer
07/18/2026  04:31 PM    <DIR>          .spyder-py3
10/08/2026  12:37 PM    <DIR>          .VirtualBox
02/04/2025  10:51 AM    <DIR>          .vscode
07/29/2026  05:48 PM    <DIR>          .vscode-shared
01/29/2026  02:34 PM                 0 30
02/14/2025  02:27 PM    <DIR>          anaconda_projects
10/21/2024  01:24 PM    <DIR>          Cisco Packet Tracer 8.2.2
10/18/2024  05:21 AM    <DIR>          Contacts
10/08/2026  12:11 PM                20 CPE11S4CAMBEL2
07/30/2026  10:47 AM    <DIR>          Desktop
10/07/2026  05:48 PM    <DIR>          Documents
10/08/2026  11:34 AM    <DIR>          Downloads
04/07/2025  02:21 PM    <DIR>          Favorites
10/22/2025  05:58 PM                 6 jcbd.txt
10/22/2025  05:56 PM                 2 jcd.txt
02/05/2025  03:29 PM    <DIR>          Jedi
10/22/2025  06:13 PM    <DIR>          journals
10/18/2024  05:21 AM    <DIR>          Links
10/22/2025  05:53 PM                 2 m.txt
10/07/2026  05:04 PM    <DIR>          Microsoft
10/18/2024  05:21 AM    <DIR>          Music
10/08/2026  10:30 AM    <DIR>          OneDrive
01/20/2025  03:16 PM    <DIR>          Pictures
10/18/2024  05:21 AM    <DIR>          Saved Games
10/17/2024  03:58 PM    <DIR>          Searches
02/14/2025  02:28 PM                72 Untitled.ipynb
02/14/2025  02:29 PM                72 Untitled1.ipynb
03/21/2025  02:42 PM             2,297 Untitled10.ipynb
04/04/2025  02:28 PM             1,913 Untitled11.ipynb
04/25/2025  03:03 PM               628 Untitled12.ipynb
07/18/2026  02:55 PM               337 Untitled13.ipynb
02/14/2025  02:53 PM             2,711 Untitled2.ipynb
02/14/2025  03:11 PM                72 Untitled3.ipynb
02/14/2025  03:16 PM             3,904 Untitled4.ipynb
02/28/2025  02:32 PM               628 Untitled5.ipynb
02/28/2025  03:28 PM                72 Untitled6.ipynb
03/07/2025  03:33 PM               804 Untitled7.ipynb
03/07/2025  03:58 PM             2,334 Untitled8.ipynb
03/07/2025  04:00 PM             2,989 Untitled9.ipynb
08/06/2025  05:04 PM    <DIR>          venv
10/21/2024  01:20 PM    <DIR>          Videos
10/08/2026  11:23 AM    <DIR>          VirtualBox VMs
              21 File(s)         19,185 bytes
              36 Dir(s)  504,912,760,832 bytes free
```

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e. Use the appropriate commands to delete all the text files and folders that you have created in this lab. Type **exit** to close the command prompt window.

Reflection Question

What are the advantages of using CLI vs. GUI?

The Command Line Interface (CLI) provides key advantages such as faster execution speed for experienced users, lower resource consumption, powerful automation and scripting capabilities, and efficient remote management options over low-bandwidth connections. In contrast, the Graphical User Interface (GUI) excels in user-friendliness, visual intuitiveness, and ease of navigation, making it ideal for beginners and casual users who prefer non-textual interaction.

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